

return np.floor(np.sqrt(x))

→ Bounding down

$$26. \sum_{j=0}^m \lfloor \sqrt[3]{j} \rfloor$$

$$\text{floor}(3.2) = 3$$

$$\text{floor}(2) = 2 = \lfloor 2 \rfloor \rightarrow \text{Returns same integer}$$

| j  | $\sqrt[3]{j}$ | $\lfloor \sqrt[3]{j} \rfloor$ |
|----|---------------|-------------------------------|
| 0  | 0             | 0                             |
| 1  | 1             | 1                             |
| 2  | 1.3           | 1                             |
| 3  | 1.4           | 1                             |
| 4  | 1.6           | 1                             |
| 5  | 1.7           | 1                             |
| 6  | 1.9           | 1                             |
| 7  | 2             | 2                             |
| 8  | 2.1           | 2                             |
| 9  | 2.3           | 2                             |
| 10 | 2.4           | 2                             |
| 11 | 2.6           | 2                             |
| 12 | 2.7           | 2                             |
| 13 | 2.9           | 2                             |
| 14 | 3             | 3                             |
| 15 | 3.1           | 3                             |
| 16 | 3.3           | 3                             |
| 17 | 3.4           | 3                             |
| 18 | 3.6           | 3                             |
| 19 | 3.7           | 3                             |
| 20 | 3.9           | 3                             |
| 21 | 4             | 4                             |

$$2^3 = 8$$

$$1 \leq x < 8$$

$$\lfloor \sqrt[3]{x} \rfloor = 1$$

$$0 + 1 + 1 + 1 + 1 + 1 + 1 + 2 + \dots + 2 + 3 + \dots + 4 + \dots$$

$$7 \cdot 1 = 7 \cdot 1$$

$$26 - 8 + 1 = 17 \cdot 2$$

$$63 - 27 + 1 = 37 \cdot 3$$

$$(k+1)^3 - 1$$

$$\sum_{k=1}^n \sum_{j=k^3}^{(k+1)^3-1} \lfloor \sqrt[3]{j} \rfloor$$

$$\text{For all } k^3 \leq x \leq (k+1)^3 - 1$$

Suppose

$$m = (n+1)^3 - 1$$

K doesn't depend on j

$$\lfloor \sqrt[3]{x} \rfloor = k$$

$$37 = 64 - 27$$

$$\sum_{k=1}^n \sum_{j=k^3}^{(k+1)^3-1} k = \sum_{k=1}^n k \sum_{j=k^3}^{(k+1)^3-1} 1$$

$$= \sum_{k=1}^n k((k+1)^3 - k^3) \rightarrow \text{next pg.}$$